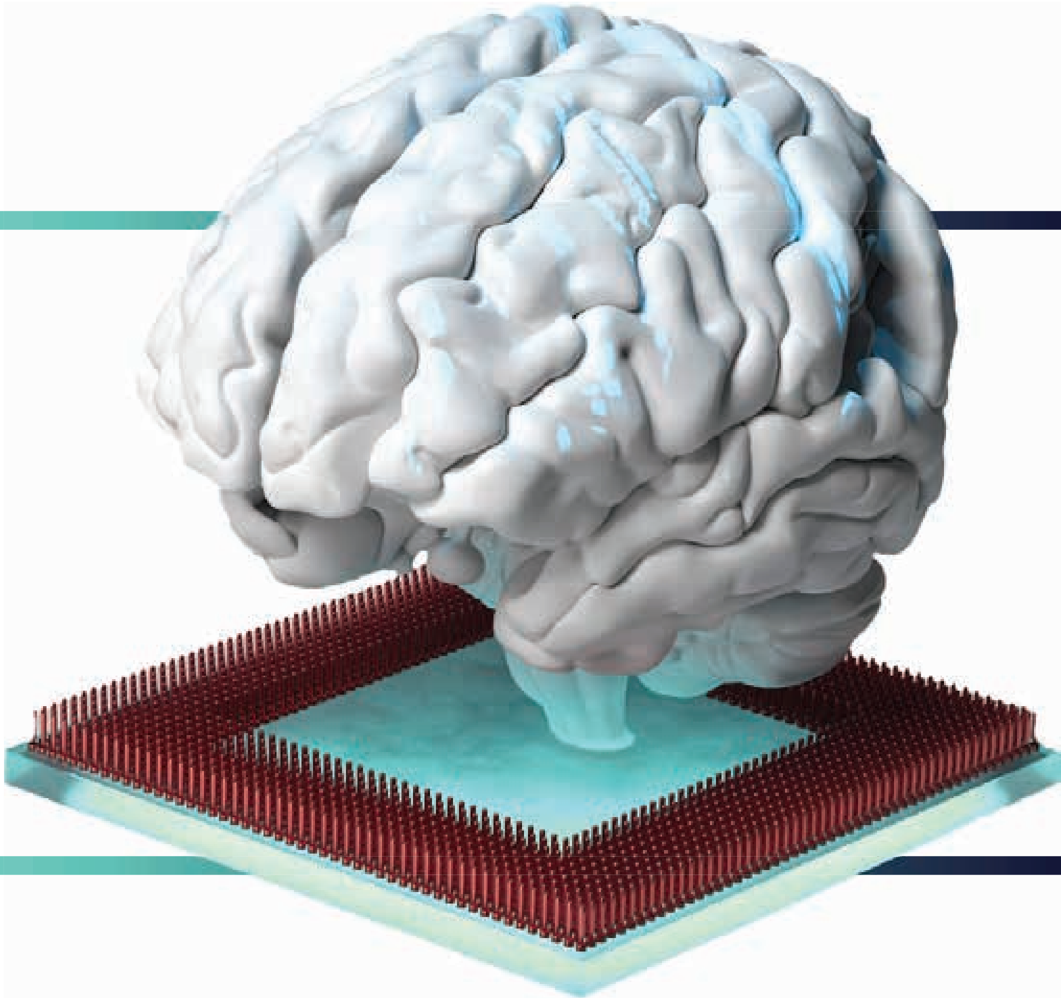
INDIAN
INSTITUTE OF
TECHNOLOGY
ROORKEE

Batch 11



Post Graduate Certificate in

Data Science, Machine Learning & Generative AI

- 8 Months
- Live Online + Campus Immersion

Programme offered through the Continuing Education Centre, IIT Roorkee

AI, ML & Data Science: **Powering the Next Decade of Digital Transformation**

In today's digital economy, Data Science and Artificial Intelligence are no longer niche skills — they are the engines driving every modern business decision. From healthcare to finance, retail to manufacturing, organisations are shifting from intuition-led to data-led decision making, powered by automation, advanced analytics, and now, Generative AI.

What was once limited to predictive models has rapidly evolved into intelligent systems that can reason, create, converse, and automate end-to-end workflows. With the convergence of Machine Learning, MLOps, Generative AI and Data Engineering, the industry is moving towards AI agents, autonomous analytics pipelines, and real-time decisioning at scale.

According to global forecasts, the AI and Data Science market is expected to cross USD 1 trillion by 2030, with GenAI alone contributing over 30% to enterprise productivity gains. This wave of transformation has triggered an unprecedented demand for professionals who can not only build ML models, but also deploy, scale, and integrate AI into business applications.

Programme **Overview**

The Post Graduate Certificate in Data Science, Machine Learning & Generative AI by IIT Roorkee is a hands-on, industry-ready programme designed for professionals looking to build full-stack AI expertise. Covering everything from Python, ML, Deep Learning, and MLOps to LLMs and Generative AI, the programme offers live sessions, real-world projects, and a capstone for practical application. With two specialisation tracks — Deep Learning with GenAI or Data Engineering with GenAI — learners can tailor their career path toward AI development or deployment roles. Supported by career guidance and mentorship, the programme prepares participants to transition into high-demand roles such as Data Scientist, ML Engineer, or Data Engineer.

Programme **Highlights**

Premier certification from Indian Institute of Technology Roorkee (Continuing Education Centre)

Comprehensive 8-month curriculum covering Data Science, ML Deep Learning, MLOps & Generative AI

Live online sessions by IIT Roorkee faculty & industry experts

Hands-on learning with 10+ industry-relevant projects + Capstone

Two Specialisation Tracks to choose from:

- Deep Learning with Image & Speech using Generative AI
- Data Engineering with Generative AI (Azure, Spark, Kafka, ETL)

MLOps & Deployment Skills – CI/CD pipelines, Docker, MLFlow

Training on cutting-edge AI tools

Optional 5-day Campus Immersion at IIT Roorkee for in-person exposure & networking

Weekend-friendly format for working professionals

Note: Campus immersion expenses are the responsibility of the participant.

Tools
You Will Learn

- databricks
- Spark SQL
- Azure
- python
- Keras
- OpenCV
- NLTK
- mongoDB
- GitHub Actions
- Spark MLlib
- Spark
- matplotlib
- seaborn
- tableau
- docker
- scikit learn
- Dialogflow
- TensorFlow
- Transformers (BERT/GPT)
- LangChain

Key
Learning Outcomes

Upon completing the programme, learners will be able to:

- Build strong foundations in Data Science and Python-based analytics
- Work with industry-standard data science libraries - NumPy, Pandas, TensorFlow, Scikit-learn, Matplotlib.
- Analyse, clean, and visualise data to derive meaningful insights
- Develop Machine Learning models for prediction, classification, and clustering
- Build Deep Learning models for advanced AI applications
- Apply Computer Vision techniques with OpenCV, CNNs, and pretrained models for image classification, object detection, and face recognition.
- Implement speech recognition and advanced NLP workflows using embeddings, RNNs, LSTMs, and transformers.
- Understand and apply Generative AI and LLMs in real-world scenarios
- Design AI systems using prompt engineering, RAG, LangChain, and AI agents
- Understand the foundations of LLMs, Diffusion Models, GANs, Word2Vec, BERT, and GPT.
- Build and manage end-to-end ML pipelines with CI/CD, Docker, and GitHub Actions.
- Deploy and monitor production-ready AI/ML models
- Work with Big Data and Data Engineering tools (Azure, Spark, Kafka, Databricks)
- Work with Azure Data Lake, ADLS Gen2, Databricks, Apache Spark, Kafka, and NoSQL databases.
- Solve real-world business problems through hands-on projects.

Programme Curriculum

Module 1 - Orientation

Sub-Modules

- Technical and academic orientation

Module 2 - Foundations of Data Science

Sub-Modules

- Introduction to Gen AI
- Introduction to ML, AI, Bias & Variance, prediction and inference models

Module 3 - Python for data science

Sub-Modules

- Introduction to Python
- Data structures
- Conditional statements, loops and functions
- Numpy, Pandas
- Scipy, Matplolib

Module 4 - Exploratory Data Analysis and Data Visualization

Sub-Modules

- Data extraction and cleaning
- Data visualization using Python
- Data visualization using Tableau
- Regular expressions
- RDBMS and SQL

Module 5 - Machine Learning

Sub-Modules

- ML Project cycle and Naïve Bayes classifier with Hands-on
- Regression analysis
- Logistic regression; Softmax regression; One hot encoding
- SVM and KNN classifiers
- Clustering approaches
- Decision Trees

Module 6 - Advanced Machine Learning

Sub-Modules

- Ensemble learning: bagging and boosting models
- Linear and Non-linear dimensional reduction techniques
- Statistical/Probabilistic and Density methods for clustering
- Gaussian Mixture models for clustering and data generation

Module 7 - Text Analytics

Sub-Modules

- Basics of text analytics
- Text pre-processing, frequency based analysis
- Converting text to numbers, building ML models from text data
- N-gram models, evaluating language models
- Word2Vec, applications of deep models to text data

Module 8 - Neural Networks & Deep Learning

Sub-Modules

- Introduction to ANN
- ANN through Scikitlearn
- Introduction to Keras and Tensorflow
- Sequential and Functional models
- Improving the performance of neural networks

Module 9 - MLOps

Sub-Modules

- Version Control Systems
- Containerization (Docker)
- CI/CD Pipelines GitHub Actions
- CI/CD pipeline: AWS
- CI/CD pipeline: Deployment

Module 10 - Generative AI

Sub-Modules

- Prompt fundamentals
- Building GenAI tools/custom GPT
- Prompt techniques

Module 11
Specialization 1 -
Deep Learning with Image and Speech with Generative AI

Sub-Modules

- Image: Introduction to computer vision, open CV, CNN
- Image: Introduction and Hands-on popular CNN architectures
- Image: Popular object detection architectures
- Image: Autoencoders, GANs, Diffusion models
- Speech: Processing sequential data using RNN
- Speech: Processing sequential data using LSTM
- Speech: Introduction to word embeddings and advanced NLP
- Introduction to Transformers architecture
- GenAI: RAG
- GenAI: AI agents and multi-agent systems
- GenAI: LangChain

Module 11
Specialisation 2 -
Data Engineering with Generative AI

Sub-Modules

- Introduction to Big Data and HDFS, Azure Data Lake Storage Gen2 & Blob, Distributed systems, replication & lifecycle management
- Apache Spark & RDD architecture, PySpark APIs, Spark SQL & data frame, aggregations & window functions, ETL pipelines
- Batch v/s Streaming architecture, Structured streaming in Spark, Kafka fundamentals, streaming data frames
- Feature engineering with Spark, MLlib for feature pipelines
- ETL/ELT with Azure data factory, linked services & pipelines, triggers, serverless transformations
- AI-assisted data cleaning & transformations, schema/metadata generation, automated feature enrichment, missing value imputation, AI-powered ETL
- Complete Azure-based Data Engineering pipeline with Gen AI integration

Capstone Project

Note: Curriculum matrix subject to change based on emerging industry trends and academic requirements.

Top Job Roles
You May Explore

Data Scientist
Designs and builds predictive models to solve business problems using data.

Machine Learning Engineer
Develops and deploys ML algorithms into scalable production systems.

Data Analyst / Business Analyst
Interprets data insights and helps stakeholders make data-driven decisions.

AI / Generative AI Engineer
Builds intelligent systems using LLMs, chatbots, and GenAI automation tools.

Deep Learning Engineer
Works on advanced neural networks for computer vision, NLP, and speech applications.

Data Engineer
Builds and manages data pipelines and infrastructure for large-scale analytics.

MLOps Engineer
Streamlines ML workflows using CI/CD and Docker for model deployment.

Decision Scientist / Applied AI Specialist
Connects AI outputs with business strategy to drive measurable outcomes.

Big Data Engineer
Handles high-volume data processing using tools like Spark, Kafka, and Azure Data Lake.

AI Automation & LLM Engineer
Creates AI agents and workflow automations powered by large language models.

Programme offered through the Continuing Education Centre, IIT Roorkee

Programme

Details & Eligibility

Eligibility

- Bachelor's degree with minimum 50%
- Learners with prior background in Engineering, Technology, Computer Science, IT, Mathematical Sciences and related disciplines will be preferred.
- Minimum 1 years of experience preferably in IT, software, technology or engineering domain

8 months

- 122 hours of instructor led live online sessions
- 130 hours of live session for students (one specialization)
- 196 hours of recorded videos, assignments, projects, etc.
- 30 hours of campus immersion
- 30 hours of capstone project

Class Schedule - Every Saturday and Sunday: 9am to 11am

Admission Criteria

Selection based on application review

Campus Immersion

5 days (30 hours) of campus immersion at the end of the programme (optional for the learners to attend).

Pedagogy - Theory % - 50% | Practical % - 50%

Note: Session timings are tentative and subject to revision.

Assignments/Case-studies/Projects/

- IPL Analytics eBay Car Sales Analytics E-commerce Customer Shopping Analytics
- Classification of Human Activity Recognition
- Predictive model to forecast the sales of supermarket
- Customer Segmentation of Clickstream Online Retail Shopping Data
- Popularity Prediction of Social Media Articles
- Bitcoin Price Prediction
- Time Series Analysis of Energy Consumptions of Appliances
- Ensemble Techniques to Classify the Customer Churn
- Regularization Techniques for IPL Auction Analysis
- Story telling with Netflix Text Data
- Topic Modelling on Amazon Review Dataset
- Personality Classification based on MBTI Metric
- Fake News Detection
- Auto tagging of photos uploaded by the users on review website
- COVID-19 Detection Using Chest X Rays

Assessment and Evaluation Criteria

- 60% - Assignments & Project
- 20% - Capstone Project
- 10% - Quizzes
- 10% - attendance
- Learners need to complete all the modules with at least one assignment submission and a minimum 60% quiz score.

Attendance
Criteria

- A minimum of 75% attendance is required for successful completion of the IIT Roorkee Data Science and Machine Learning programme.
- Full attendance is recommended for in-depth understanding since several modules will be covered during a span of the programme.

Certificate
Criteria

Candidates who score at least 50% marks overall and have a minimum attendance of 75%, will receive a 'Certificate of Completion'.

Career Support

Note: Career Support Facility is provided by TimesPro. IIT Roorkee is not responsible for the same.

Personal Branding

- Introduction to networking platforms
- Profile creation on professional networking platforms like LinkedIn, Lunchclub, etc.
- LinkedIn Profile Review
- How to create personal brand presence on LinkedIn?
- How to increase post engagement on LinkedIn?
- Active networking

Business Communication

- Role and importance of effective communication as a leader
- Mastering the art of delivering constructive feedback for cultivating successful team dynamics.
- Importance of non-verbal communication
- Key elements of executive body language

Career Advancement Strategy

Resume Creation

- Importance of creating ATS friendly executive resume
- Executive resume sections and structure
- Tailoring resumes for different roles and industries
- Write a powerful resume that stands out from the competition
- Resume Review - Peer to peer review and Q&A

Interview Preparation

Pre-interview Etiquettes

- Learn about top-down approach for interviews
- Pre-interview tips and tricks

In-interview Etiquettes

- Create a self-elevator pitch
- Understanding interviewer mindset
- Interview grooming sessions and tips and tricks for interview

Post-interview Etiquettes

- Reflecting on interview experience and incorporating the feedback
- Relationship building with the recruiter
- Learn how to follow up on your job application

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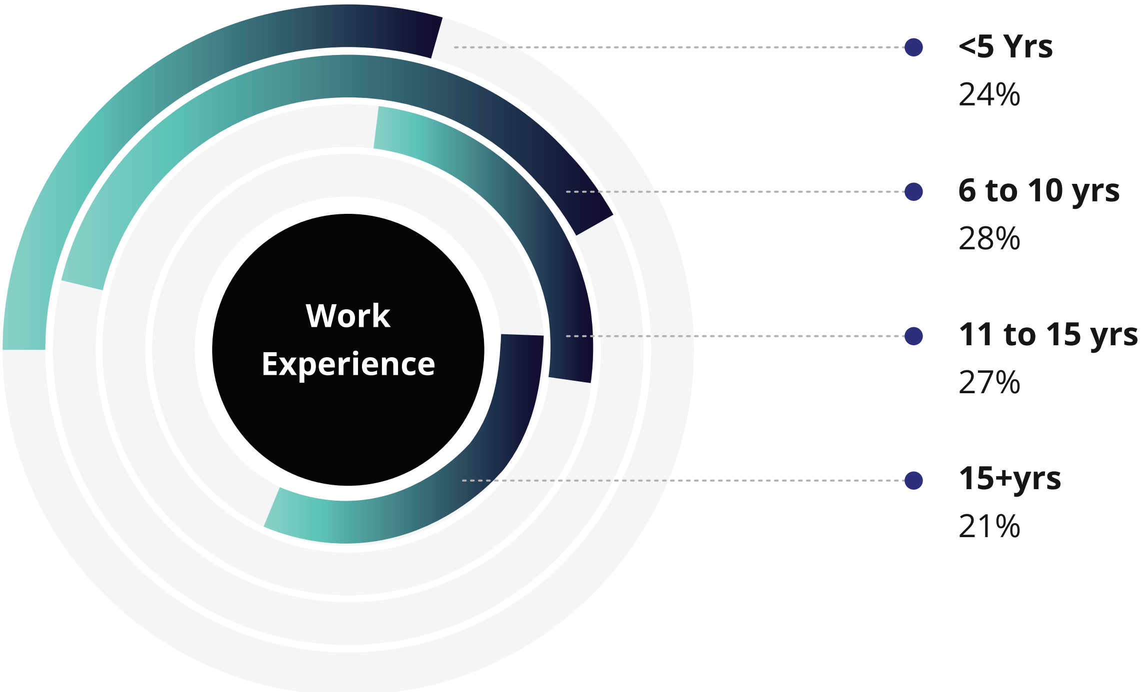
Campus
Immersion

Past Participant profile



Industries

22%
Information Technology

17%
Computer Software

14%
Finance

5%
Engineering

41%
Others

*Others include Automotive, Telecommunications, Market Research, Graphic Design, Health Care, Management Consulting, amongst others.

Programme
Coordinator

Prof. Alok Bhardwaj

Assistant Professor,

Department of Civil Engineering and
Mehta Family School of Data Science and Artificial Intelligence,

Indian Institute of Technology, Roorkee

Prof. Alok Bhardwaj's research interests include the application of deep learning, computer vision, and digital image processing to earth observation datasets. Dr. Bhardwaj is a Joint Faculty at the Mehta Family School of Data Science and Artificial Intelligence and is also a National Geographic Explorer.

VIEW PROFILE

Prof Sanjeev Kumar Malik

Professor,

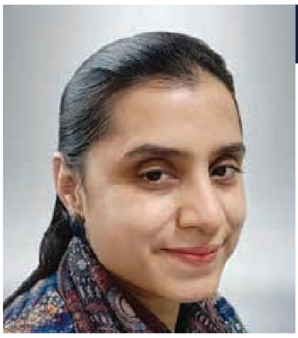
Department of Mathematics, IIT Roorkee.

Prof. Sanjeev Kumar works in the area of mathematical image processing including computational algorithms for image restoration, image encryption and secret sharing, Quantum Imaging and 3-D Reconstruction, machine learning, and applications in image processing.

VIEW PROFILE

Programme

Coordinator



Prof. Millie Pant

Professor,

Department of Applied Science and Engineering, IIT Roorkee.

Prof. Millie Pant has been associated with IIT Roorkee since 2007. Her areas of interest include Numerical Optimisation, Operations Research and Supply Chain Management, among others.

VIEW PROFILE

Prof. Sumit Kumar Yadav

Assistant Professor,

Operations Management, IIT Roorkee

Prof. Sumit Kumar Yadav completed his Ph.D. in Management from IIM Ahmedabad and B.Tech from IIT Bombay. His teaching experience spans IIT Roorkee and reputed management colleges.

VIEW PROFILE

Programme
Coordinator

Prof. Gaurav Kumar Nayak

Assistant Professor,

Mehta Family School of Data Science and Artificial Intelligence

Professor Gaurav Kumar Nayak is an Assistant Professor at IIT Roorkee, specialising in Generative AI, Data-efficient Deep Learning, and Computer Vision. A former Post-doctoral Scholar at UCF and IISc Best PhD Thesis Award recipient, he currently leads high-impact research projects funded by global organisations like Cisco and Graylark Technologies.

VIEW PROFILE

Programme

Timelines & Fee Structure

Application Fees- ₹1,000 (non-refundable)

Total Fee (Exclusive of Application Fee) - ₹1,75,000

Note:

*GST @18% will be charged extra in addition to the fee

Instalment Schedule

Component	Date	Amount ₹* (Exclusive of GST)
Application fee	To be paid at the time of Application	₹1000
Installment- 1	within 7 days of the offer letter or before the start of the program, whichever is earlier.	₹43,750
Installment- 2	28th September, 2026	₹43,750
Installment- 3	28th October, 2026	₹43,750
Installment- 4	27th November, 2026	₹43,750

Last Date To Apply	**Refer To Website**
Programme Start date	**29th August, 2026**
Batch End	**26th April, 2027**

APPLY NOW

Complimentary Access to

3 Certificate Programmes

As part of this programme, you get exclusive access to 3 industry-relevant courses designed to strengthen your practical AI & GenAI capabilities

01 Quantum Computing: Algorithms & Applications

Build a foundational understanding of quantum computing and how it integrates with modern AI systems.

- Learn quantum fundamentals, algorithms & real-world use cases
- Understand the evolution of computing and quantum ecosystems
- Explore applications in security, communication & advanced computing
- Learn to apply mathematical concepts relevant to quantum computing

02 Certificate Program in Agentic AI: Unlocking Generative AI Potential

Learn how to design AI systems that can plan, reason, and execute tasks autonomously.

- Build AI agents & intelligent workflows
- Gain experience with platforms like Auto-GPT, LangChain & CrewAI
- Deploy low-code / no-code AI agents
- Apply agentic AI in real-world business scenarios

03 Certificate Program in GenAI Tools & Business Applications

Understand how Generative AI is applied across business functions to drive real impact.

- Hands-on tool exposure via demos and walk-throughs
- Understand the principles of ethical and responsible AI implementation
- Apply AI across HR, finance, operations
- Learn AI strategy, ethics & ROI-driven adoption

What You Get?

- Learn at your own pace alongside the main programme
- Gain hands-on exposure across emerging AI domains
- Build a well-rounded, application-focused AI skillset

Disclaimer: The three complimentary Certificate Programmes are offered by TimesPro. IIT Roorkee is not responsible for the same. Certificates for the complimentary programmes will be issued by TimesPro upon successful completion of the Post Graduate Certificate Programme in Data Science, Machine Learning & Generative AI.

INDIAN INSTITUTE OF TECHNOLOGY ROORKEE

The University of Roorkee, now the Indian Institute of Technology, had a modest beginning as the Thomason College of Engineering in the year 1847. It was converted into the First Technical University of India in 1949 and into an IIT in 2001. IIT Roorkee, a highly ranked institution, is among the foremost institutes of national importance in higher technological education and in engineering, basic and applied research. It is also a trend-setter in the area of education and research in the field of science, technology, and engineering.

The institute offers Bachelor's Degree in 10 disciplines of Engineering and Architecture and Postgraduate Degree in 55 disciplines of Engineering, Applied Science, Architecture and Planning. The institute has facility for doctoral work in all Departments and Research Centres.

About CEC IIT Roorkee

Continuing Education Center (CEC) IIT Roorkee was established in 1955 for the promotion of knowledge up-gradation by organising refresher/specialist courses for executives, working professionals and aspiring individuals. In addition to the knowledge up-gradation of experienced professionals, CEC IIT Roorkee courses also offer a launchpad to young technology professionals and students in the domains that are most sought after by the industry. Apart from the technical expertise available in the Departments and Centres of the Institute, wherever necessary, experts from industries and R&D organisations are also invited to deliver talks/lectures. CEC conducts sponsored and open participation programmes in various disciplines of engineering, science, technology and management for learners both nationally and globally. Through these courses, CEC also provides the opportunity to visit and study at IIT Roorkee for a short duration.

QS World University Rankings, 2026

#339
Global University Rankings

#114
Asia University Rankings

NIRF 2025

#7
Overall

#6
Among Engineering Colleges

#1
Architecture and Planning

India's Best Architecture College 2025, India Today

#1
Department of Architecture and Planning, IIT Roorkee

Times Higher Education World University Rankings, 2024

#501-600
World University Rankings

#83
Asian University Rankings

Established in 2013, we are the award-winning H.EdTech initiative of the Times Group, catering to the learning needs of Indians with aspirations of career growth. We offer a variety of created and curated learning programmes across a range of categories, industries, and age groups. They include employment-oriented Early Career courses across BFSI, e-Commerce, and technology sectors; Executive Education for working professionals in collaboration with premier national and global educational institutions; and Enterprise Solutions for learning and development interventions at the organisational level. TimesPro strives to embody the values of Education 4.0: Learner-centric, industry-relevant, role-specific, and technology-enabled, with a goal of making learning accessible for anyone who seeks to grow.

Industry relevant curriculum by best-in-class faculty

Interactive sessions with state-of-art LMS

IIMs and IITs as course partners

1,00,000+ alumni community

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